



Original research

Examination of the external and internal load indicators' association with overuse injuries in professional soccer players



Arne Jaspers^{a,*}, Jurian P. Kuyvenhoven^b, Filip Staes^c, Wouter G.P. Frencken^b,
Werner F. Helsen^a, Michel S. Brink^b

^a KU Leuven, Department of Kinesiology, Movement Control and Neuroplasticity Research Group, Belgium

^b Center for Human Movement Sciences, University of Groningen, University Medical Center, The Netherlands

^c KU Leuven, Department of Rehabilitation Sciences, Musculoskeletal Rehabilitation Research Group, Belgium

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ABSTRACT

Objectives: Research in professional soccer focusing on the relevance of external and internal load indicators for injury prevention is scarce. This study examined the relationship between load indicators and overuse injuries.

Design: Prospective cohort study.

Methods: Data were collected from 35 professional male soccer players over two seasons. Following load indicators were examined: total distance covered (TD), distance covered at high speed (THSR; $>20 \text{ km h}^{-1}$), number of accelerations (ACC_{eff} ; $>1 \text{ m s}^{-2}$), number of decelerations (DEC_{eff} ; $<-1 \text{ m s}^{-2}$), and rating of perceived exertion (RPE) multiplied by duration. Cumulative 1-, 2-, 3-, 4-weekly loads and acute:chronic workload ratios (ACWR) were calculated and split into low, medium and high groups. Only overuse injuries were included in the analysis to focus on their specific relationship with the load indicators. Generalized estimating equations were applied to analyse the relationship between load indicators and overuse injuries in the subsequent week.

Results: In total, 64 overuse injuries were registered. For cumulative loads, results indicated an increased injury risk for higher 2- to 4-weekly loads as indicated by TD, DEC_{eff} , and RPE multiplied by duration. For ACWR, a high ratio for THSR (>1.18) resulted in a higher injury risk. In contrast, a lower injury risk was found when comparing medium ratios for ACC_{eff} (0.87–1.12), DEC_{eff} (0.86–1.12), and RPE x duration (0.85–1.12) to low ratios.

Conclusions: Findings demonstrate that mainly external load indicators are associated with increased or decreased injury risk. The monitoring of various load indicators is recommended for injury prevention in professional soccer.

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1. Introduction

Professional soccer players sustain on average 2.0 injuries per season, which cause them to miss 37 days in a 300-day season on average.¹ Training and match load are considered to be strongly associated with injuries, however, these loads were not included in previous injury aetiology models.² Following the updated injury aetiology model, training and match load contribute together with intrinsic and extrinsic risk factors to the multifactorial and dynamic aetiology of injury.² Not only excessive loading and insufficient

recovery, but also underpreparedness may increase injury risk by exposing players to large relative changes, or spikes, in load during periods with higher training and match loads.³ These spikes can be identified using the acute:chronic workload ratio (ACWR).⁴ Therefore, training and match load monitoring is considered essential to optimize load management and to minimize injury risk.⁴

Training and match load are generally quantified in terms of external and internal loads.⁵ The external load refers to all player's locomotor movements and can be measured using electronic tracking systems such as global positioning systems (GPS) and accelerometers. The external load is quantified in terms of distance, velocity and accelerations.⁴ The internal load refers to the physiological response of players to external load and can be determined using heart rate (HR) and ratings of perceived exertion (RPE).⁴ The relationship with overuse injury for both external and

* Corresponding author at: University of Leuven (KU Leuven), Department of Kinesiology, Tervuursevest 101 (box 1501), B-3001 Leuven, Belgium.

E-mail address: arne.jaspers@kuleuven.be (A. Jaspers).

internal load indicators has been examined in various elite team sports such as Australian football, cricket, rugby league,⁴ and also in youth soccer.⁶ In professional soccer, however, little evidence exists with respect to the load indicators that may be related to injury.⁷

For internal load, one study investigated the relationship between RPE multiplied by duration and injury risk.⁸ One study found a relationship between HR and overuse injuries, in particular a positive correlation between muscular strains and training intensity measured by average HR.⁹ Aforementioned studies focusing on internal load already stated the need to consider GPS data (i.e., external load indicators) to examine the relationship between injuries and external load indicators, especially in terms of high-intensive activities such as high-speed running, accelerations and decelerations.^{8,9}

To date, one study in professional soccer found a relationship between non-contact soft tissue injuries and a higher distance covered per minute in the weeks before injury, in comparison with the players' season average values.¹⁰ Additionally, lower average values for an external load indicator based on triaxial accelerometry, when compared to players' season average, were found in the weeks before injury.¹⁰ No relationship was found for injury risk and distance covered at high speed.¹⁰ High-speed running is related to non-contact soft tissue injuries in other team sports.^{11–13} However, one limitation of the study by Ehrmann et al.¹⁰ is the use of 5 Hz GPS units. This sampling frequency exhibited limitations in terms of accuracy and reliability when applied for measuring high-intensity efforts.¹⁴ These limitations may have impacted on the results for high-speed running variables.¹⁰ Additionally, accelerations and decelerations were not examined by the authors due to the 5 Hz sampling frequency limitations.¹⁰

In professional soccer, these high-intensive activities are considered important to monitor.¹⁵ Interestingly, accelerations, decelerations and RPE multiplied by duration have not been studied yet for their relationship with overuse injury risk.⁷ Assessment of this relationship may provide evidence for their implementation and succeeding predictive research to optimize load management strategies in professional soccer.¹⁶ Therefore, the aim of the present study was to examine different external and internal load indicators in relation to overuse injuries.

2. Methods

Thirty-five professional male soccer players (mean $\pm$ SD age: 23.2 ± 3.7 years, weight: 77.5 ± 7.4 kg, height: 1.82 ± 0.06 m, body fat: $10.4 \pm 1.9\%$) participated in this study. They were all players of the first team competing at the highest level in the Netherlands (Eredivisie). Goalkeepers were not included. Data were collected over two seasons (2014–2015 and 2015–2016), including pre-season and in-season. Written informed consent was obtained according to the Helsinki declaration. The study was approved by the ethical committee of KU Leuven (file number: s57732).

External load was quantified individually during all field training sessions and matches using 10 Hz GPS technology (Minimax S4 and Optimeye S5, Catapult Sports, Melbourne, Australia). This sampling rate has proven a good validity and reliability for high intensive movement demands.¹⁴ The data collection was completed following the guidelines for collecting and processing GPS data in sport.¹⁷ The selected external load indicators were total distance covered (TD), distance covered at high speed (THSR; >20 km h⁻¹), the number of acceleration efforts >1 m s⁻² (ACC_{eff}) and deceleration efforts <-1 m s⁻² (DEC_{eff}). For a 10 Hz sampling rate, the accuracy of higher accelerations (>4 m s⁻²) is compromised.¹⁴ Therefore, we have chosen to detect total efforts >1 m s⁻² or <-1 m s⁻². The minimum effort duration to detect veloc-

ity was 0.6 s, and 0.4 s for acceleration with a smoothing filter of 0.2 s.^{17,18}

Following each field training session and match, data was downloaded using the manufacturer's software (Catapult Sprint, 5.1.7), checked for irregularities (i.e., spikes in velocity data), satellite connection (≥ 8 satellites), and horizontal dilution of precision (<1.5), and then processed.¹⁷ If data quality requirements were not met or player data were missing, values were estimated following Bowen et al.⁶ For field training sessions, values were estimated for individual players using the average of players with a similar position that took part in the same training session ($n = 193$ of 6536; 3%). In addition, match data for season 2014–2015 were estimated due to FIFA restrictions regarding the use of GPS units in official matches. Therefore, match values were estimated by means of the player's average based upon measured data of friendly games and matches during season 2015–2016. Playing time was taken into account for all match value estimations.^{6,10,13} For season 2015–2016, data was collected during all matches and only estimated if data quality requirements were not met ($n = 121$ of 873; 14%).¹⁷ In total, the number of estimated external load data for field training sessions and matches was 870 of 8103 (11%).

Internal load was obtained for each individual following gym sessions, field training sessions and matches using RPE scores using the modified Borg CR-10 scale.¹⁹ The RPE was administered approximately 30 min after the end of training sessions or matches to ensure that the perceived intensity would reflect the session as a whole.¹⁹ All athletes were familiarized with the scale before the start of the study. The load in arbitrary units (AU) was derived for each player by multiplying RPE with training or match duration.

Injuries were diagnosed and recorded by members of the medical staff. The data collection procedures were in accordance with the consensus statement for soccer injury studies.²⁰ An injury classification system was embedded within a medical data management system to code each diagnosis by location, type, and mechanism of injury. All injuries during both seasons were recorded, but only time-loss overuse injuries that resulted in a player being unable to take a full part in soccer training or match play were included in the analyses.²⁰ An overuse injury is defined as an injury caused by repeated micro-trauma without a single, identifiable event responsible for the injury.²⁰ Injury incidence was calculated by dividing the number of injuries by exposure time and reported as rate per 1000 training and match hours.²⁰

Data were categorized into weekly blocks from Monday until Sunday. Weeks in which players were away with national teams were excluded from further analyses ($n = 177$ of 1764; 10%). Cumulative 1-, 2-, 3-, 4-weekly loads were calculated as the sum of the daily load of the previous week(s).²¹ The ACWR was calculated weekly by dividing the 1-week load of the most recent week by the 4-week rolling average weekly load.²² These load variables were calculated for selected external and internal load indicators.

Data were analyzed using SPSS version 24 (IBM Corporation, New York, USA). Generalized estimating equations (GEE) were used to model the univariate association between each load variable and overuse injuries in the subsequent week.²³ The model was set for a binary distribution of the dependent variable (injury yes/no), logit link function, first-order autoregressive (AR1) working correlation structure, player as subject variable, weeks as within-subject variable and all load variables were modelled independently as predictor variable.

GEE was used for its ability to provide a population averaged effect from repeatedly measured data of multiple subjects. Data were first tested for normality and randomization of missing values. Load variables were sorted from lowest to highest and split into tertiles to divide the data in low, medium and high load groups. The lowest load group served then as reference group to compare injury risk with medium and high load group and allowed for non-

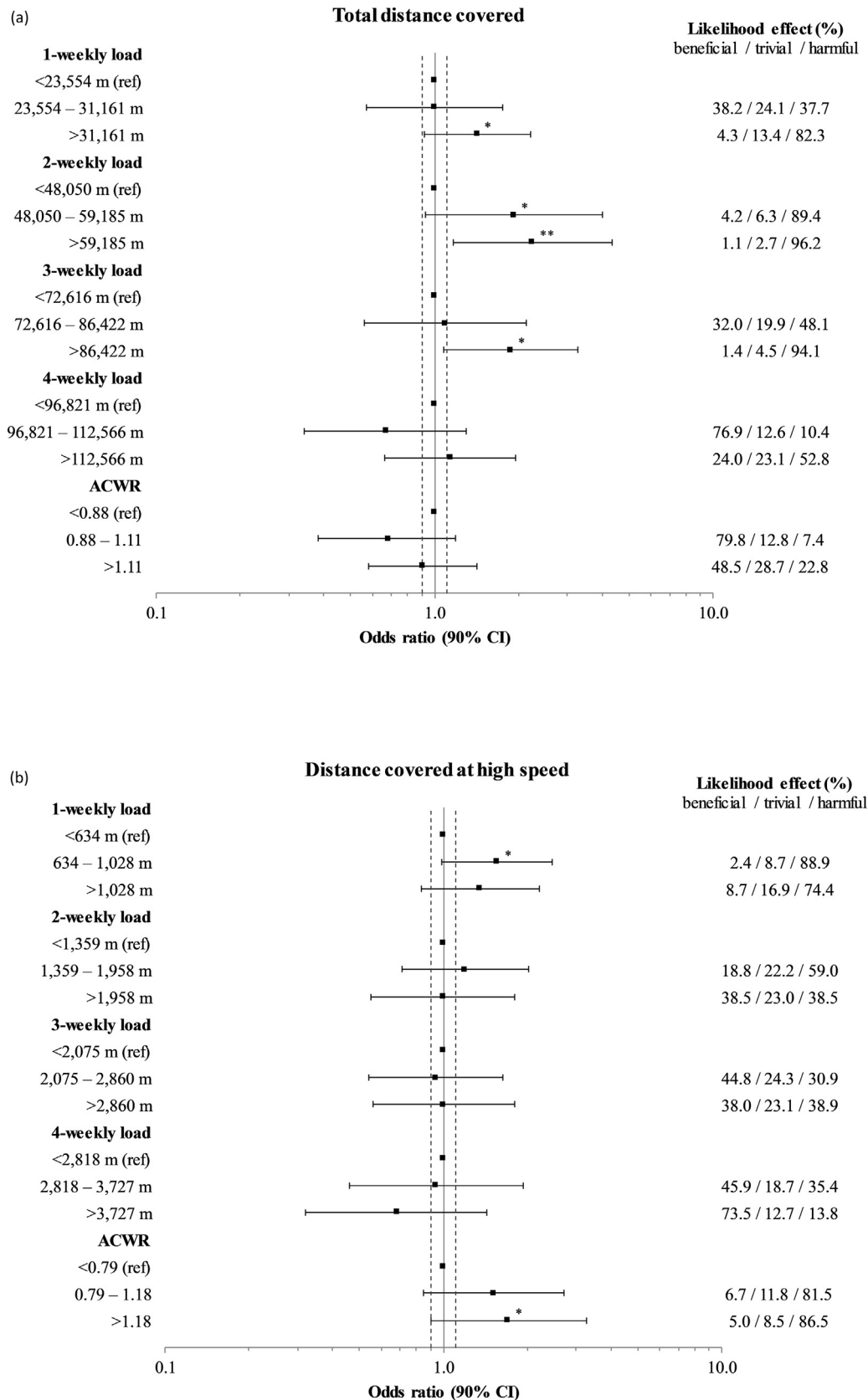


Fig. 1. Injury risk of TD (m) and THSR (m) for cumulative 1-, 2-, 3-, 4-weekly loads and ACWR at medium and high loads compared to low loads (reference group (ref)). Smallest worthwhile changes in injury risk are displayed as dotted lines. Likelihood of effects are shown in percentages. † = Likely beneficial, †† = very likely beneficial, ††† = most likely beneficial, * = likely harmful, ** = very likely harmful, *** = most likely harmful.

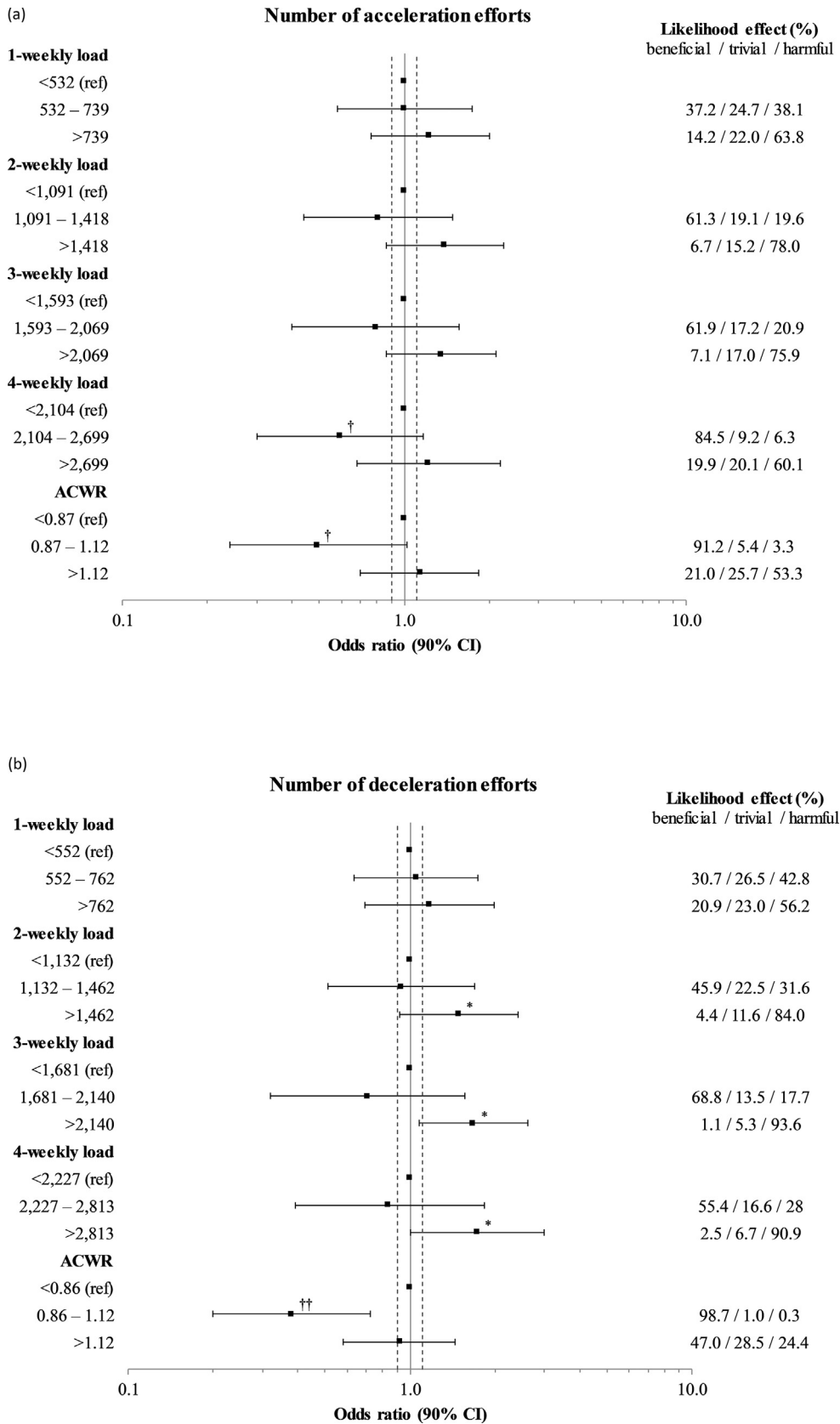


Fig. 2. Injury risk of ACC_{eff} (number of efforts) and DEC_{eff} (number of efforts) for cumulative 1-, 2-, 3-, 4-weekly loads and ACWR at medium and high loads compared to low loads (reference group (ref)). Smallest worthwhile changes in injury risk are displayed as dotted lines. Likelihood of effects are shown in percentages. † = Likely beneficial, †† = very likely beneficial, ††† = most likely beneficial, * = likely harmful, ** = very likely harmful, *** = most likely harmful.

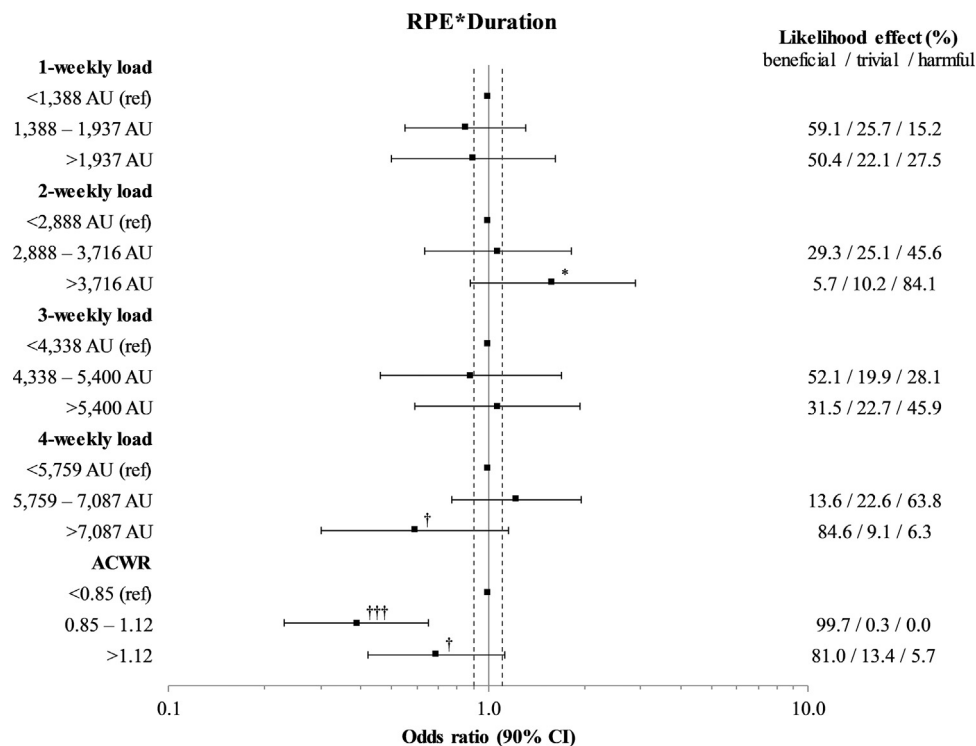


Fig. 3. Injury risk of RPE x duration (AU) for cumulative 1-, 2-, 3-, 4-weekly loads and ACWR at medium and high loads compared to low loads (reference group (ref)). Smallest worthwhile changes in injury risk are displayed as dotted lines. Likelihood of effects are shown in percentages. † = Likely beneficial, †† = very likely beneficial, ††† = most likely beneficial, * = likely harmful, ** = very likely harmful, *** = most likely harmful.

linear analyses of predictors on the outcome variable.²⁴ Odds ratios (OR) and 90% confidence intervals (CI) were calculated to determine which load variables increased or decreased the risk of injury.

Magnitude-based inferences were used to provide an interpretation of the real-world relevance of the outcome.²⁵ Effects were reported in quantitative and qualitative terms using the following scale: <0.5%, most unlikely; 0.5–5%, very unlikely; 5–25%, unlikely; 25–75%, possible; 75–95%, likely; 95–99.5%, very likely; >99.5%, most likely.²⁶ Practical significance was reached when the probability that the true effect was either harmful or beneficial was $\geq 75\%$, likely.

3. Results

In total, 64 time-loss overuse injuries were included in the study, respectively 27 in the first season and 37 in the second season. The overuse injury incidence over two seasons was 5.8 injuries per 1000 h, respectively 4.9 injuries per 1000 h for the first season and 6.8 injuries per 1000 h for the second season.

The association between load and overuse injury in the subsequent week using TD, THSR, ACC_{eff} , DEC_{eff} , and RPE x duration are shown in Figs. 1–3, respectively. An overview is provided of harmful and beneficial effects in terms of injury risk for medium and high loads in comparison with low loads (reference group) for 1-, 2-, 3-, 4-weekly loads and ACWR.

Results are presented in order of harmful or beneficial effect. Only most likely, very likely and likely effects are described. The description of these effects is structured from external to internal load indicators, and from cumulative 1-, 2-, 3-, 4-weekly loads to ACWR.

No most likely harmful effects were found. A very likely harmful effect was found for a high 2-weekly TD (>59,185 m, OR: 2.25, 90% CI: 1.17–4.34). Likely harmful effects were found for TD: for a high 1-weekly TD (>31,161 m, OR: 1.42, 90% CI: 0.92–2.21), for a medium

2-weekly TD (48,050–59,185 m, OR: 1.93, 90% CI: 0.93–4.02), and for a high 3-weekly TD (>86,422 m, OR: 1.88, 90% CI: 1.08–3.26). Also, likely harmful effects were observed for THSR: for a medium 1-weekly THSR (634–1028 m, OR: 1.56, 90% CI: 0.99–2.46), and for a high ACWR for THSR (>1.18, OR: 1.71, 90% CI: 0.90–3.26). Likely harmful effects were found for DEC_{eff} : for a high 2-weekly DEC_{eff} (>1462, OR: 1.49, 90% CI: 0.92–2.42), for a high 3-weekly DEC_{eff} (>2140, OR: 1.68, 90% CI: 1.08–2.63), and for a high 4-weekly DEC_{eff} (>2813, OR: 1.73, 90% CI: 1.00–2.99). Finally, a likely harmful effect was observed for RPE x duration: for a high 2-weekly RPE x duration (>3716, OR: 1.59, 90% CI: 0.88–2.89).

A most likely beneficial effect was found for a medium ACWR for RPE x duration (0.85–1.12, OR: 0.39, 90% CI: 0.23–0.65). A very likely beneficial effect was found for a medium ACWR for DEC_{eff} (0.86–1.12, OR: 0.38, 90% CI: 0.20–0.72). Likely beneficial effects were found for ACC_{eff} : for a medium 4-weekly ACC_{eff} (2104–2699, OR: 0.59, 90% CI: 0.30–1.17), and for a medium ACWR for ACC_{eff} (0.87–1.12, OR: 0.49, 90% CI: 0.24–1.02). Also, likely beneficial effects were found for RPE x duration: for a medium 4-weekly RPE x duration (>7087, OR: 0.59, 90% CI: 0.30–1.16), and for a high ACWR for RPE x duration (>1.12, OR: 0.69, 90% CI: 0.42–1.13).

4. Discussion

The aim of this study was to examine the relationship between load indicators and overuse injuries in professional soccer. Higher cumulative 1-, 2-, 3-weekly loads for TD and 2-, 3-, 4-weekly loads for DEC_{eff} were associated with increased injury risk in the subsequent week. In addition, a high ACWR for THSR was related to a higher injury risk, while a medium ACWR for ACC_{eff} , DEC_{eff} , and RPE x duration was linked to a lower injury risk in the subsequent week.

Harmful effects were found for high weekly loads of TD. In Australian football, increased 3-weekly total distance was likewise very

highly indicative of greater risk for intrinsic (i.e., overuse) injuries.¹³ This possibly indicates the relevance of monitoring an overall external volume indicator for managing injury risk.

For THSR, a medium 1-weekly load and a high ACWR were related to an increased injury risk. Small differences in high-speed running or sprinting can influence injury risk.^{11,12} Duhig et al.¹² found that large and rapid increases in individualised THSR ($>24 \text{ km h}^{-1}$) should be avoided to prevent hamstring injuries in Australian football. Moreover, they found no relationship with injury risk for cumulative weekly loads of THSR. These findings are in line with our results and probably indicate the relevance of the use of relative and individualized THSR thresholds to examine the relationship with injury risk.²⁷

None of the weekly loads for ACC_{eff} was related to a higher injury risk. However, for DEC_{eff} , higher 2-, 3-, 4-weekly loads were related to an increased injury risk. The sport-specific activities performed during soccer contain numerous eccentric muscle contractions that induce muscle damage.²⁸ Consequently, muscles need to recover after training sessions and matches, if not, injury can occur due to a mismatch between load and recovery.²⁸ Therefore, absolute amounts of DEC_{eff} could be relevant to monitor.

For RPE multiplied by duration, only a high 2-weekly load exhibited a higher injury risk. RPE multiplied by duration has been found a good method to quantify internal load for resistance training.²⁹ Therefore, also gym sessions were quantified, next to training sessions and matches, in line with earlier research in professional team sports.³⁰ This method, quantifying load of all sessions, does not exhibit higher odds ratios for the estimation of injury risk in comparison with external load indicators for training sessions and matches only. Although some weaknesses concerning RPE to monitor injury risk have already been addressed,²⁷ it is worth mentioning that RPE for matches may be influenced by match-related parameters such as location, opponent's level and result.^{31,32} Furthermore, some issues were highlighted regarding the terminology of effort or exertion and the sensitivity of scales used to determine RPE.³³ Because of the complex interaction of factors contributing to RPE, monitoring differential RPE focusing on cardiovascular, musculoskeletal and cognitive exertion may further improve the interpretation of internal load and possible associations with injury risk.³⁴

In general, cumulative 2- to 4-weekly loads demonstrated higher ORs for TD and DEC_{eff} . This may indicate that high cumulative 1-weekly loads do not significantly increase to injury risk, however, when high loads are maintained for longer periods injury risk may increase substantially. Only for THSR, injury risk already increases for higher 1-weekly loads.

Beneficial effects were found for the load indicators ACC_{eff} , DEC_{eff} , and RPE x duration. A lower injury risk was found for a medium 4-weekly load for ACC_{eff} and a high 4-weekly load for RPE x duration. These results confirm the possible protective effects of medium weekly loads in comparison with lower loads.³⁰ A lower overuse injury risk was also found in Australian footballers with a higher 1-weekly load for RPE x duration in comparison with the lower load group.³⁰

Beneficial effects were exhibited for medium ACWR for ACC_{eff} , DEC_{eff} , and RPE x duration, showing a decreased injury risk in the subsequent week. This is in line with earlier research in different team sports indicating the importance of a gradual load distribution and preparedness to manage injury risk.⁴ For RPE x duration, a relationship was found between a decreased injury risk and both medium and high ACWR. The rather low threshold (>1.10) could possibly explain the lower injury risk for the high ACWR. In a professional soccer study using the same RPE method, a lower injury risk was found for ACWR values between 1.00 and 1.25.⁸

This was the first study in professional soccer to examine both external and internal load indicators in relation to overuse injury risk.⁷ Several insights are provided regarding the relevance of these indicators to improve monitoring strategies for the prevention of overuse injury. The applied research at the highest level in professional soccer is a strength of the current study. However, the lower sample size, inherent to research in professional soccer, limits the possible statistical analyses and the immediate applicability to other teams. Furthermore, some additional limitations of this single-team study need to be considered.

According to the multifactorial aetiology of injury, internal risk factors (e.g., physical fitness and injury history) and external risk factors (e.g., equipment and environment) may affect injury risk.² This study focused on the univariate relationship between load indicators and overuse injury risk and did not include internal or external risk factors. In addition, it is important to consider that this is a study of association and not of prediction.¹⁶ Despite this specific focus, several relationships were found between load indicators and both higher and lower injury risks. These associations may provide evidence to support their implementation in load monitoring strategies.¹⁶ In future research, the inclusion of internal and external risk factors can further improve the estimation of injury risk and possibly allow for predictive studies. Nevertheless, this is only possible for larger datasets including more professional teams over a period of multiple seasons.³⁵ Additionally, the inclusion of the players' response to match and training load, using subjective self-reported measures,³⁶ could further clarify the process of overuse injuries.³⁷

Limitations in the use of ACWR in professional soccer are international duties and off-season periods.²⁷ For these periods, ACWR was unavailable and data was excluded from analysis in this study. Additionally, ACWR needs further research regarding daily ACWR instead of weekly ACWR, the acute:chronic time window, and injury lag period.³⁸ Analogous to data unavailability during international duties, missing player data and the subsequent estimation of external load indicators are a limitation in the current study. However, missing data for training sessions and matches are considered inherent to applied research within elite team sports as exhibited and handled in earlier research.^{6,10,13}

5. Conclusion

The results of the current study show that mainly external load indicators are associated with increased or decreased injury risk. The monitoring of cumulative weekly loads for TD and DEC_{eff} is advised. Additionally, a high ACWR for THSR should be avoided. On the contrary, also protective effects were found and a medium ACWR is recommended for ACC_{eff} , DEC_{eff} , and RPE x duration. In conclusion, this study demonstrates a delicate balance for various external and internal load indicators regarding increased and decreased injury risk. This emphasises the relevance to monitor these variables closely for optimizing load management and reducing injury risk in professional soccer.

Practical implications

- The findings of this study provide recommendations about the relevance of monitoring external and internal load indicators in professional soccer to reduce risk for overuse injury.
- Both cumulative weekly loads and acute:chronic workload ratios should be monitored to optimize day-to-day load management.
- External load indicators may be more relevant to monitor than RPE multiplied by duration to minimize injury risk.

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